Mandatory Assignment 2

GFS189, RTL959

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In this assignment we examine whether machine learning can be used to predict stock returns and generate economically meaningful investment strategies. The analysis is inspired by Gu et al. (2020), who show using a broad set of machine learning models that non-linear methods outperform simple historical-mean benchmarks for cross-sectional return prediction. This stands in contrast to Welch and Goyal (2007), who find that machine learning models generally fail to beat the historical mean for aggregate market return prediction. We attempt to replicate part of the Gu et al. (2020) findings using their stock characteristics and macroeconomic predictors.

1 Data Preparation

In this section we load and prepare the stock-level dataset for return prediction. The data are stored in a parquet file containing 94 stock characteristics from Gu et al. (2020) and 13 macroeconomic variables from Welch and Goyal (2007), along with industry classifications, excess returns, and lagged market capitalizations.

We select ten stock characteristics and four macroeconomic variables based on figure 5 form the paper of Gu et al. (2020). In figure 1 we see the distribution of the stock characterstics. This shows that some of the variables are non-normal and skewed highlighting the usefullness of cross-sectional median imputation and winsorisation. Figure 2 shows the monthly excess return by SIC division. All divison except "other" showcases a positive mean monthly excess return.

variable	count	mean	std	min	1%	5%	10%	25%	50%	75%	90%	95%	99%	max
characteristic_mom1m	1.1116e+06	-0.004	0.604	-0.996	-0.99	-0.914	-0.827	-0.55	-0.009	0.544	0.827	0.915	0.99	0.995
characteristic_mom12m	1.1116e+06	-0.007	0.59	-0.997	-0.99	-0.909	-0.817	-0.526	-0.026	0.519	0.819	0.911	0.99	0.992
characteristic_mom36m	1.1116e+06	-0.008	0.56	-0.995	-0.99	-0.899	-0.796	-0.469	-0.026	0.458	0.797	0.9	0.99	0.992

variable	count	mean	std	min	1%	5%	10%	25%	50%	75%	90%	95%	99%	max
characteristic_bm	1.1116e+06	-0.065	0.531	-0.99	-0.979	-0.898	-0.801	-0.512	-0.067	0.373	0.684	0.796	0.899	0.99
characteristic_ep	1.1116e+06	-0.011	0.558	-0.99	-0.979	-0.898	-0.797	-0.486	-0.004	0.454	0.764	0.878	0.976	0.99
characteristic_mvel1	1.1116e+06	0.026	0.588	-1	-0.971	-0.891	-0.795	-0.496	0.044	0.544	0.824	0.916	0.985	1
characteristic_dolvol	1.1116e+06	0.035	0.597	-1	-0.98	-0.908	-0.818	-0.502	0.076	0.56	0.833	0.919	0.99	0.99
characteristic_roaq	1.1116e+06	-0.014	0.56	-0.991	-0.99	-0.903	-0.803	-0.477	-0.023	0.457	0.778	0.885	0.973	0.99
characteristic_gma	1.1116e+06	0.046	0.56	-0.99	-0.99	-0.893	-0.773	-0.416	0.073	0.512	0.808	0.903	0.979	0.99
characteristic_invest	1.1116e+06	-0.006	0.533	-0.99	-0.976	-0.882	-0.768	-0.427	-0	0.413	0.75	0.871	0.973	0.99
macro_dp	1.1116e+06	-4.022	0.208	-4.524	-4.493	-4.436	-4.346	-4.107	-4.001	-3.904	-3.845	-3.786	-3.384	-3.281
macro_ep	1.1116e+06	-3.177	0.382	-4.836	-4.807	-3.824	-3.595	-3.317	-3.113	-2.917	-2.809	-2.758	-2.663	-2.566
macro_tbl	1.1116e+06	0.018	0.019	0	0	0	0	0.001	0.011	0.028	0.049	0.057	0.061	0.062
macro_tms	1.1116e+06	0.022	0.014	-0.004	-0.003	-0	0.002	0.009	0.023	0.035	0.04	0.042	0.044	0.045

The final feature matrix contains 1,111,605 observations and 59 columns: 10 stock characteristics, 40 interaction terms (characteristics $\times$ macroeconomic variables), and 9 SIC division dummies.

2 Machine Learning Models

We train two machine learning models to forecast monthly excess returns: a random forest and a neural network. These are chosen based on their prevelance in the literature as a whole.

Random forest. The objective is to minimize mean squared error (MSE) by averaging predictions from an ensemble of independently grown decision trees, each trained on a bootstrap sample of the data and a random subset of features at each split. The key hyperparameters are: the number of trees (`n_estimators`), the maximum tree depth (`max_depth`, which controls model complexity), and the minimum number of samples required in each leaf node (`min_samples_leaf`, which acts as a regulariser).

Neural network. The objective is to minimise MSE via the Adam stochastic gradient descent algorithm. The architecture consists of fully connected layers with sigmoid activations. The key hyperparameters are: the hidden layer sizes (controlling depth and width), the L2 penalty coefficient (`alpha`, which regularises the weights), and the initial learning rate.

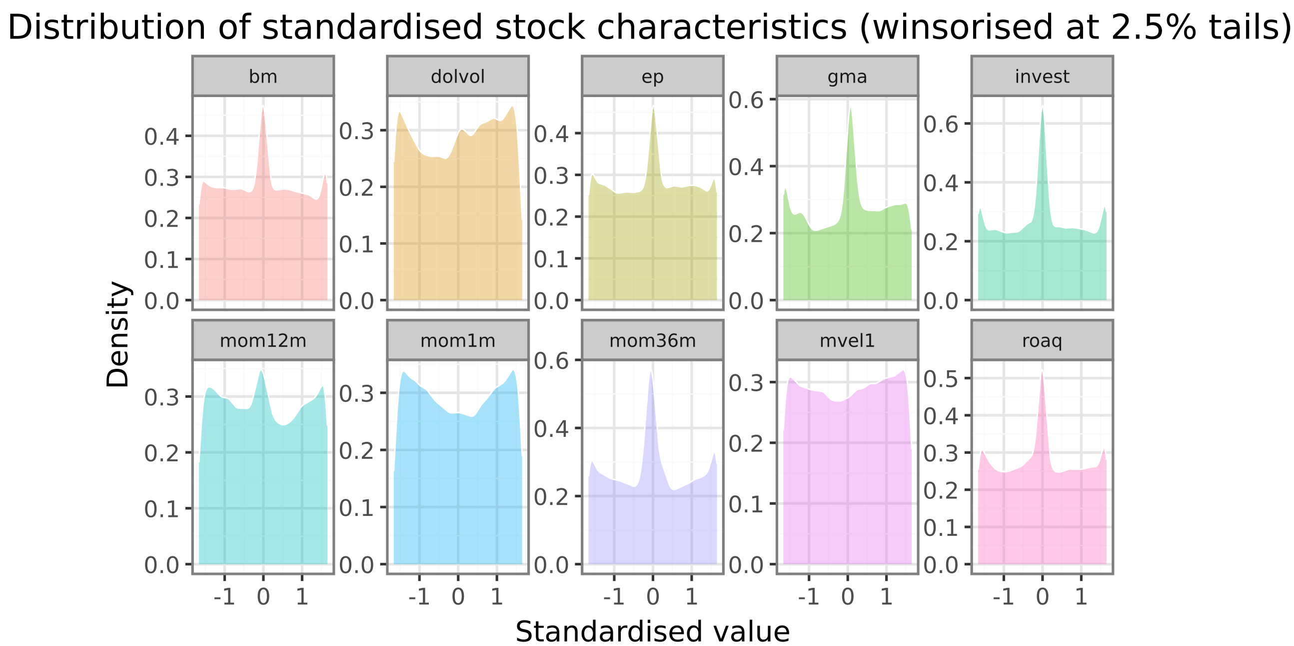

Figure 1: Distribution of standardised stock characteristics (winsorised at 2.5% each tail). Each panel shows the cross-sectional density of one characteristic after standardisation.

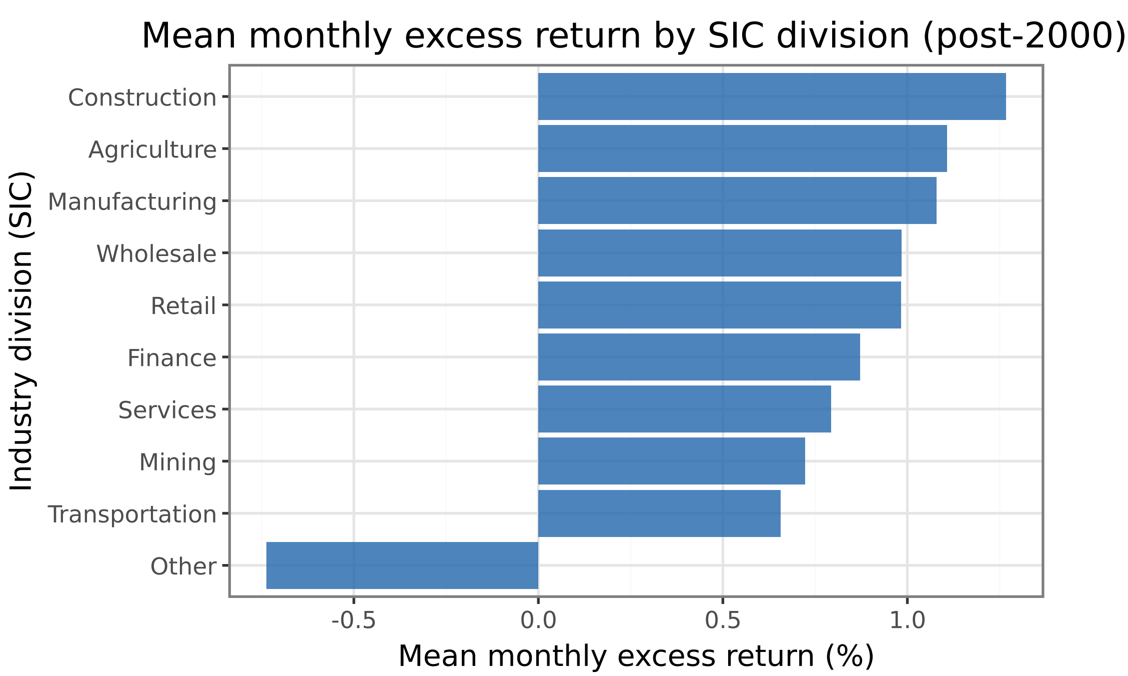

Figure 2: Mean monthly excess return by SIC division for the post-2000 sample. Bars show the time-series average of monthly value-weighted excess returns within each industry group.

Set	Start date	End date	Observations
Training	2000-01-01	2013-02-01	748,902
Validation	2013-03-01	2017-07-01	194,460
Test	2017-08-01	2021-12-01	168,243

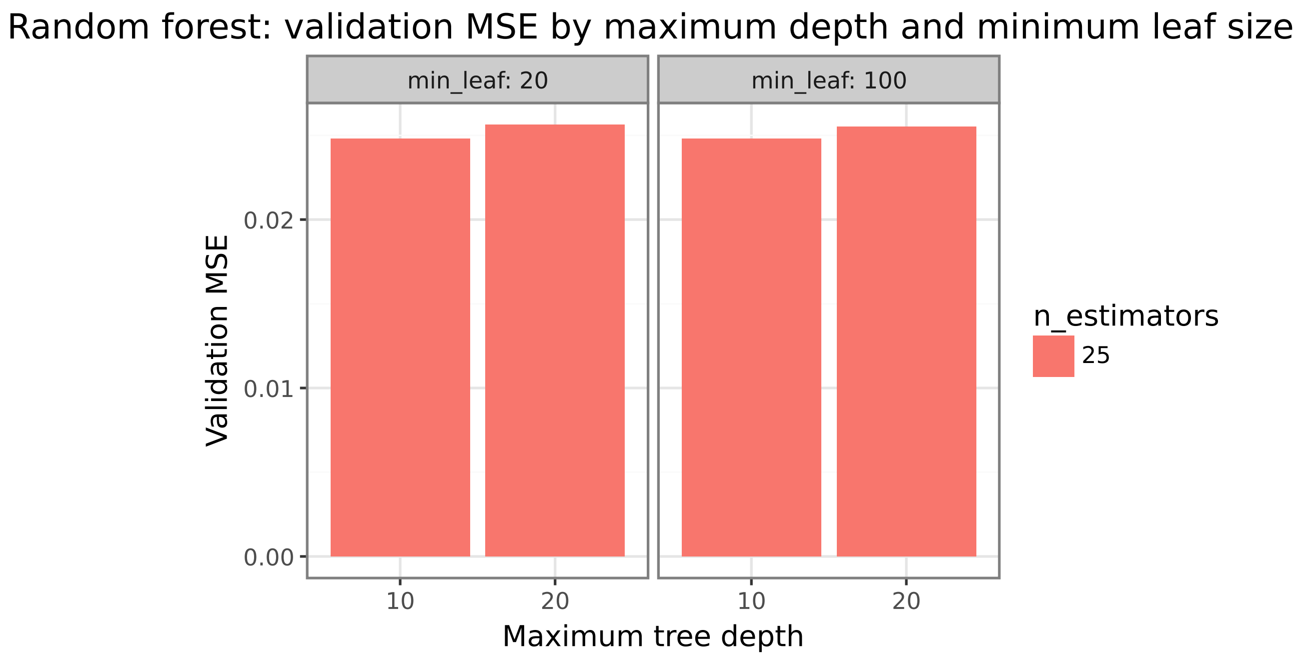

Figure 3: Random forest validation MSE by maximum tree depth and minimum leaf size. Lower MSE indicates better out-of-sample fit on the validation period.

```
['model_neural_network.joblib']
```

The fitted model objects are saved as `model_random_forest.joblib` and `model_neural_network.joblib`. A peer reviewer can load them with:

```
import joblib
best_rf = joblib.load("model_random_forest.joblib")
best_nn = joblib.load("model_neural_network.joblib")
```

The selected hyperparameters are those minimising validation MSE: for the random forest, `max_depth = 10` and `min_samples_leaf = 20`; for the neural network, architecture `(32, 16, 8)`, `alpha = 0.01`, and `learning_rate_init = 0.0005`.

The 2 figures shows tha both models achieve similar MSE around 0.022. Where the random forest is not sensitive to changing hyperparameters whereas the neural network model is more sensitivity to tuning.

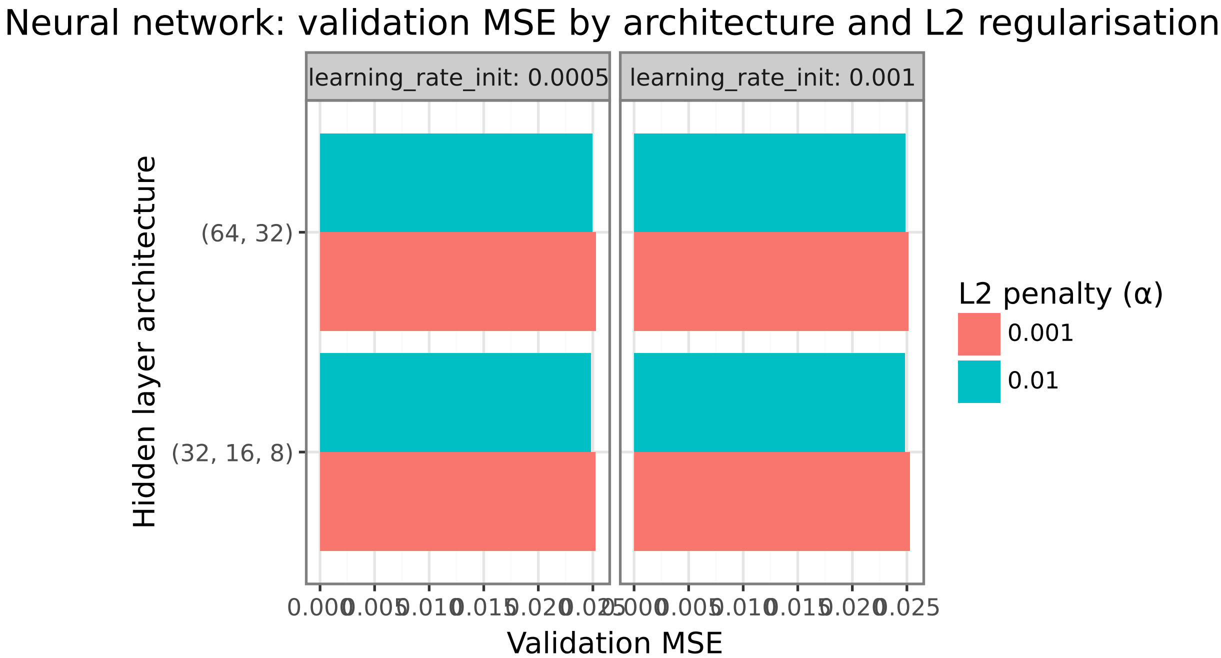

Figure 4: Neural network validation MSE by hidden layer architecture and L2 regularisation strength, faceted by initial learning rate. Lower MSE indicates better out-of-sample fit on the validation period.

3 Portfolio Construction and Performance Assessment

We translate the return forecasts into investment strategies by sorting stocks into decile portfolios each month and forming a long-short strategy.

Table 1. Long-short portfolio performance by model and period.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)	CAPM alpha (ann.)	t-statistic
RF – Validation	13.55%	0.69	12.73%	1.27
RF – Test	-10.07%	-0.37	-5.08%	-0.38
NN – Validation	-8.17%	-0.48	-11.74%	-1.35
NN – Test	-4.99%	-0.26	-7.64%	-0.8

Table 2. Market benchmark performance.

Period	Mean excess return (ann.)	Sharpe ratio (ann.)
Market – Validation	13.63%	1.33
Market – Test	17.17%	1

Table 3. Average monthly portfolio turnover by model and period.

Model	Period	Monthly turnover
RF	Validation	27.6%
RF	Test	42.7%
NN	Validation	51.8%
NN	Test	35.3%

The performance metrics in Tables 1 and 2 reveal differences between the validation and test periods. The difference in Sharpe ratios suggests that the return predictability captured in-sample does not fully persist out-of-sample, which is consistent with the difficulty of sustaining statistical arbitrage over time as market conditions change and mispricings are corrected. The turnover figures in Table 3 indicate how frequently portfolio holdings change: high monthly turnover implies substantial transaction costs that must be weighed against the gross alpha when evaluating net-of-cost profitability. As an investor neither of the models would be preferable compared to just holding the market portfolio since they both showcases negative sharpe ratios in the test and have high turnover.

4 Variable Importance and Financial Theory

Table 4. Top 3 most important predictors per model (annualised Sharpe ratio drop).

Model	Predictor	Sharpe drop
NN	dolvol	0.674
NN	mvel1	0.638
RF	tms	0.173
RF	ep	0.121
NN	mom1m	0.102
RF	mom1m	0.074

The importance rankings in Table 4 and the figure above reveal which characteristics drive portfolio performance for each model. The two models does not agree on the rankings. This reflects the models' different functional forms rather than instability in the underlying relationships.

Connecting the top predictors to financial theory, momentum variables (short-term reversal, 12-month momentum) are consistent with the well-documented momentum risk premium and the behavioral overreaction and underreaction explanations of Gu et al. (2020). Profitability variables (return on assets, gross profitability) align with the investment-based and mispricing-based explanations of the profitability factor documented by Gu et al. (2020) in their Figure 5. If valuation-based variables (book-to-market, earnings-to-price) rank highly, this supports the value premium as a compensation for systematic risk or persistent mispricing. These results are broadly consistent with Gu et al. (2020) Table 4.

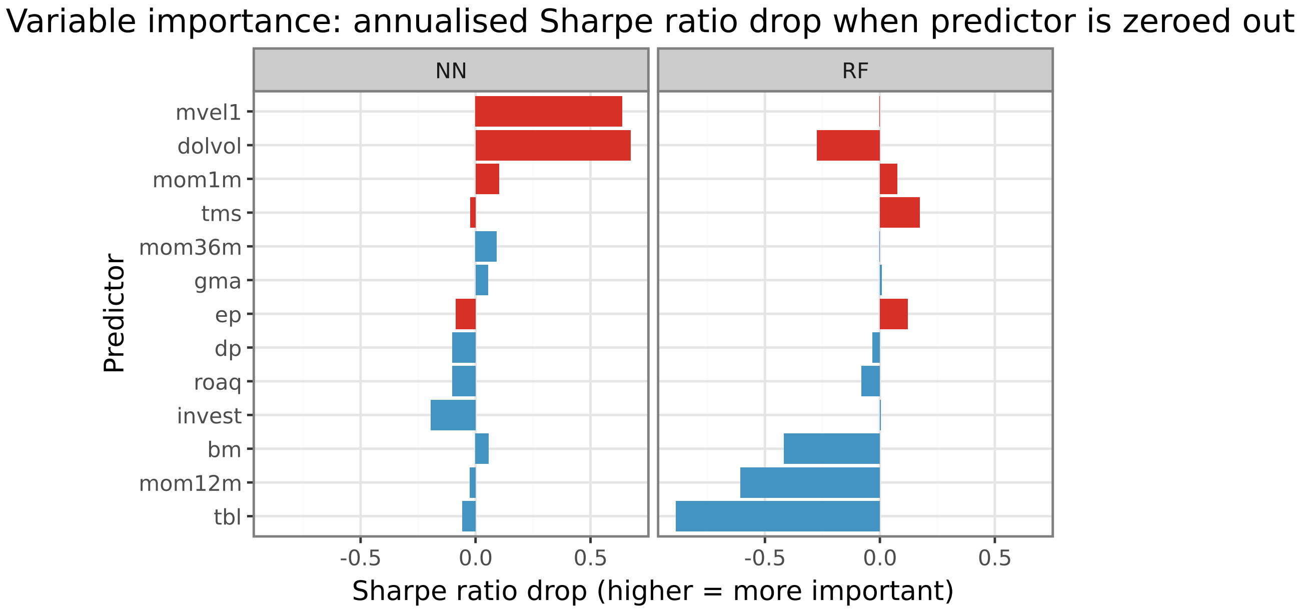

Figure 5: Variable importance measured as the drop in annualised Sharpe ratio.

5 Conclusion

This assignment replicates part of the Gu et al. (2020) framework using a random forest and a neural network trained on ten stock characteristics and four macroeconomic variables. The randon forest model generate long-short portfolio strategies with positive Sharpe ratios and CAPM alphas over the validation period. Meanwhile the neural network model strategy leads to negative sharp ratio and CAPM alpha. Performance in the test period, however, may differ from the validation period, illustrating the challenge of sustaining out-of-sample predictability. The variable importance analysis identifies momentum, profitability, and value characteristics as the primary drivers of portfolio performance, consistent with GKX's findings across a broader set of models and characteristics. High monthly turnover implies that transaction costs are material, and net-of-cost returns depend critically on the execution environment. Similar to Welch and Goyal (2007) , the results does not support the conclusion that machine learning can identify economically meaningful patterns in the cross-section of stock returns. This could be due to poor model tuning and other factors.

6 References

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.

Welch, I. and Goyal, A. (2007). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.